# How Does It Know Where to Stop?

Thermal equilibrium, energy sharing and the return of counting

**The question**

A hot copper ball and some cold water are placed together in an insulated container. Eventually both are measured at the same temperature.

**How does the system know where to stop?**

This worksheet begins with an ordinary observation and gradually changes the question. We will move through four levels:

$$\boxed{\text{observation}} \longrightarrow \boxed{\text{energy model}} \longrightarrow \boxed{\text{counting microstates}} \longrightarrow \boxed{\text{temperature}}.$$

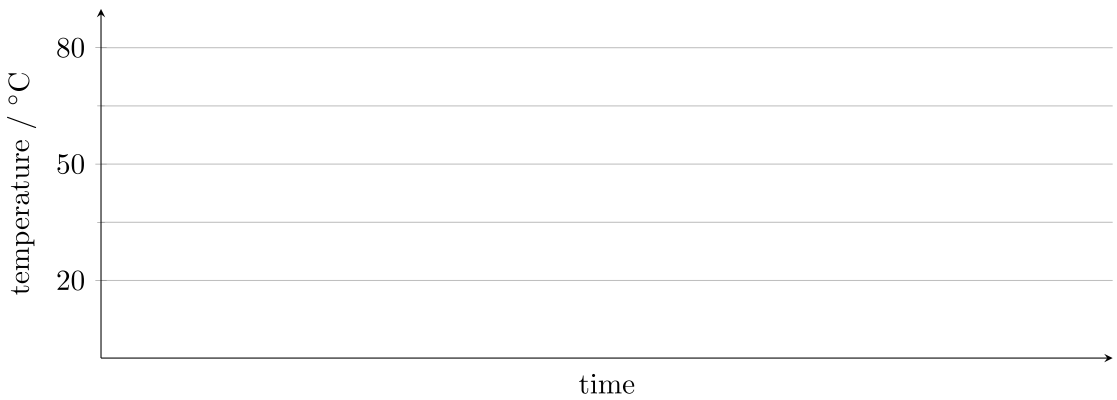
The aim is not to memorise "heat flows from hot to cold." The aim is to ask what makes that familiar behaviour overwhelmingly likely.

## 1. Sketch it before you calculate it

A copper ball begins at 80 °C. A beaker of water begins at 20 °C. They are placed together inside an insulated container.

**Think before calculating**

Without using an equation, sketch how you expect the temperature of each object to change with time.




1. Must the two curves meet? Why?

2. Must they meet at exactly 50 °C?

3. Could the curves cross so that the water temporarily becomes hotter than the copper?

4. What does it mean physically when both curves become horizontal?

## 2. What does "equilibrium" mean?

**Thermal equilibrium** is the macrostate in which there is no *systematic net transfer* of thermal energy between the objects.

That is stronger than simply saying "they have the same temperature." Equal temperature is how equilibrium is recognised from the outside. It does not yet explain why energy transfer has ceased.

5. Rewrite the definition above in your own words, without using the phrase "same temperature."

6. Why is "nothing is happening" a poor description of thermal equilibrium?

## 3. First surprise: equilibrium is not necessarily halfway

Compare two situations.

| | Case A | Case B |
|---|---|---|
| **Hot object** | 100 g water at 80 °C | 50 g copper at 80 °C |
| **Cold object** | 100 g water at 20 °C | 500 g water at 20 °C |

7. Predict the equilibrium temperature in Case A. Explain your reasoning without calculation.

8. Is 50 °C plausible in Case B? Explain.

9. What two properties of an object determine how difficult it is to change its temperature?

The heat capacity of an object is

$$C = mc,$$

where $m$ is the mass and $c$ is the specific heat capacity of the material.
A large $C$ means that a large energy transfer is needed for a given temperature change.

For an insulated two-object system,

$$\text{energy lost by hot object} = \text{energy gained by cold object}.$$

Hence

$$m_1 c_1 (T_1 - T_{\text{eq}}) = m_2 c_2 (T_{\text{eq}} - T_2),$$

so

$$\boxed{T_{\text{eq}} = \frac{m_1 c_1 T_1 + m_2 c_2 T_2}{m_1 c_1 + m_2 c_2}}.$$

Use

$$c_{\text{water}} \approx 4200\,\mathrm{J\,kg^{-1}\,K^{-1}}, \qquad c_{\text{copper}} \approx 385\,\mathrm{J\,kg^{-1}\,K^{-1}}.$$

10. Calculate $T_{\text{eq}}$ for Case A.

11. Calculate $T_{\text{eq}}$ for Case B.

12. In Case B, which matters more: that there is more water, or that water has a larger specific heat capacity than copper? Explain quantitatively.

## 4. Conservation of energy is not enough

Imagine a peculiar Tuesday.

The copper starts hotter than the water. After contact, the copper gets *hotter still*, while the water gets colder. Suppose the copper gains exactly 10 J and the water loses exactly 10 J.

**Think before calculating**

Has energy conservation been violated?

13. Show that energy can still be conserved in this strange process.

14. So why do we never expect to see this happen for ordinary macroscopic objects?

Energy conservation tells us which changes are *allowed* by the bookkeeping.
It does not by itself tell us which allowed changes are overwhelmingly likely.

## 5. Counting returns

Now replace the real copper and water by a toy model.

Each object contains a number of tiny energy stores, or oscillators. Energy comes in identical whole-number quanta. If an object has $N$ oscillators and $q$ quanta, the number of ways of distributing those quanta is

$$\boxed{\Omega(N, q) = \binom{q + N - 1}{q}}.$$

Two objects $A$ and $B$ can exchange quanta, but the total number of quanta remains fixed:

$$q_A + q_B = Q.$$

For a particular energy split,

$$\boxed{\Omega_{\text{total}} = \Omega_A(q_A)\,\Omega_B(q_B)}.$$

### A small counting experiment

Let

$$N_A = 2, \qquad N_B = 4, \qquad Q = 6.$$

Complete the table.

| $q_A$ | $q_B$ | $\Omega_A$ | $\Omega_B$ | $\Omega_{\text{total}}$ |
|---|---|---|---|---|
| 0 | 6 | | | |
| 1 | 5 | | | |
| 2 | 4 | | | |
| 3 | 3 | | | |
| 4 | 2 | | | |
| 5 | 1 | | | |
| 6 | 0 | | | |

15. Which split has the largest value of $\Omega_{\text{total}}$?

16. Did a force push the quanta toward that split?

17. Was energy conserved in all seven rows of the table?

18. Then what makes the most numerous split special?

## 6. The majority verdict

Imagine increasing the number of oscillators from a handful to something like the number of atoms in a real object. The peak in $\Omega_{\text{total}}$ becomes extraordinarily sharp.

Equilibrium is special because an overwhelming fraction of all accessible microstates correspond to macrostates extremely close to it.

A useful analogy is tossing 1000 fair coins. There is no force pushing the result toward 500 heads. There are simply enormously more sequences giving a result near 500 heads than sequences giving, say, 5 heads.

19. Explain the connection between the coin-tossing example and thermal equilibrium.

20. Why does a spontaneous flow of energy from cold water to hotter copper become effectively invisible for macroscopic objects?

## 7. Entropy enters

Boltzmann's entropy is

$$\boxed{S = k_{\mathrm{B}} \ln \Omega}.$$

The logarithm changes a product into a sum:

$$\ln(\Omega_A \Omega_B) = \ln \Omega_A + \ln \Omega_B.$$

Therefore

$$S_{\text{total}} = S_A + S_B.$$

So the most numerous energy split is also the split that maximises the total entropy.

21. Why is $\ln \Omega$ mathematically convenient when combining two systems?

22. Complete the chain:

$$\text{many microstates} \to \boxed{\phantom{xxxxxxxxxxxx}} \to \text{maximum entropy} \to \text{equilibrium}.$$

**IB extension: where equal temperature comes from**

At equilibrium, the total entropy is at a maximum.
Suppose a tiny amount of energy $dE$ is transferred from object $B$ to object $A$.
Because the total energy is fixed,

$$dE_B = -dE_A.$$

The change in total entropy is

$$dS_{\text{total}} = dS_A + dS_B.$$

Using the chain rule,

$$dS_{\text{total}} = \frac{\partial S_A}{\partial E_A}\,dE_A + \frac{\partial S_B}{\partial E_B}\,dE_B.$$

Substitute $dE_B = -dE_A$:

$$dS_{\text{total}} = \left(\frac{\partial S_A}{\partial E_A} - \frac{\partial S_B}{\partial E_B}\right) dE_A.$$

At the entropy maximum,

$$dS_{\text{total}} = 0,$$

therefore

$$\boxed{\frac{\partial S_A}{\partial E_A} = \frac{\partial S_B}{\partial E_B}}.$$

Thermodynamics defines temperature through

$$\boxed{\frac{1}{T} \equiv \frac{\partial S}{\partial E}},$$

and so the equilibrium condition becomes

$$\boxed{T_A = T_B}.$$

Thus:

$$\boxed{\Omega \longrightarrow S = k_{\text{B}} \ln \Omega \longrightarrow \frac{1}{T} = \frac{\partial S}{\partial E} \longrightarrow T_A = T_B}.$$

Equal temperature has emerged from counting and optimisation rather than being inserted at the start.

23. In words, what does $\frac{\partial S}{\partial E}$ measure?

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______________________________________________

______________________________________________

24. Why does equality of $\frac{\partial S}{\partial E}$ correspond to equilibrium?

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______________________________________________

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25. Suppose

$$\frac{\partial S_A}{\partial E_A} > \frac{\partial S_B}{\partial E_B}.$$

For which direction of a small energy transfer would $S_{\text{total}}$ increase?

26. Explain why temperature can be understood as a property describing how the multiplicity of a system responds to added energy.

## 8. The reframing

Return to the original copper ball and water.

| | |
|---|---|
| **Observed** | One stable temperature after the copper and water settle: the macrostate. |
| **Inferred** | Vast numbers of microscopic energy arrangements, overwhelmingly concentrated around energy shares corresponding to that macrostate. |

**The question**

Equilibrium is not a state in which the microscopic world has stopped.
It is a state in which almost everything the microscopic world could be doing looks macroscopically the same.

## 9. Exit reflection

Choose **one** question and answer it carefully.

- What did we assume without noticing?
- Why is conservation of energy insufficient to explain the direction of thermal change?
- What does probability add to the explanation?
- In what sense is temperature a bridge between microscopic and macroscopic descriptions?
- What would convince you that the "entropy as disorder" description is inadequate?

## Postamble: what this worksheet has changed

At the beginning, thermal equilibrium looked like a simple fact:

$$\text{hot object} + \text{cold object} \longrightarrow \text{same temperature.}$$

We can now tell a more precise story:

$$\boxed{\begin{array}{c} \text{energy is conserved} \\ \Downarrow \\ \text{many energy shares are possible} \\ \Downarrow \\ \text{their multiplicities are very unequal} \\ \Downarrow \\ \text{one narrow region dominates} \\ \Downarrow \\ S_{\text{total}} \text{ is maximal} \\ \Downarrow \\ T_A = T_B \end{array}}$$

The system does not "know" where to stop. Nothing pushes it toward a preferred temperature. The equilibrium macrostate is stable because almost all the accessible microscopic arrangements look like it from the outside.

**Physics begins with the observation.**
**The model tells us what is conserved.**
**Counting tells us what is typical.**